

Pokémon Red benchmark: Jev alone vs Jev + GPT-6 Sol

28 September 2026 · runs tagged t1 · sampling temperature 1

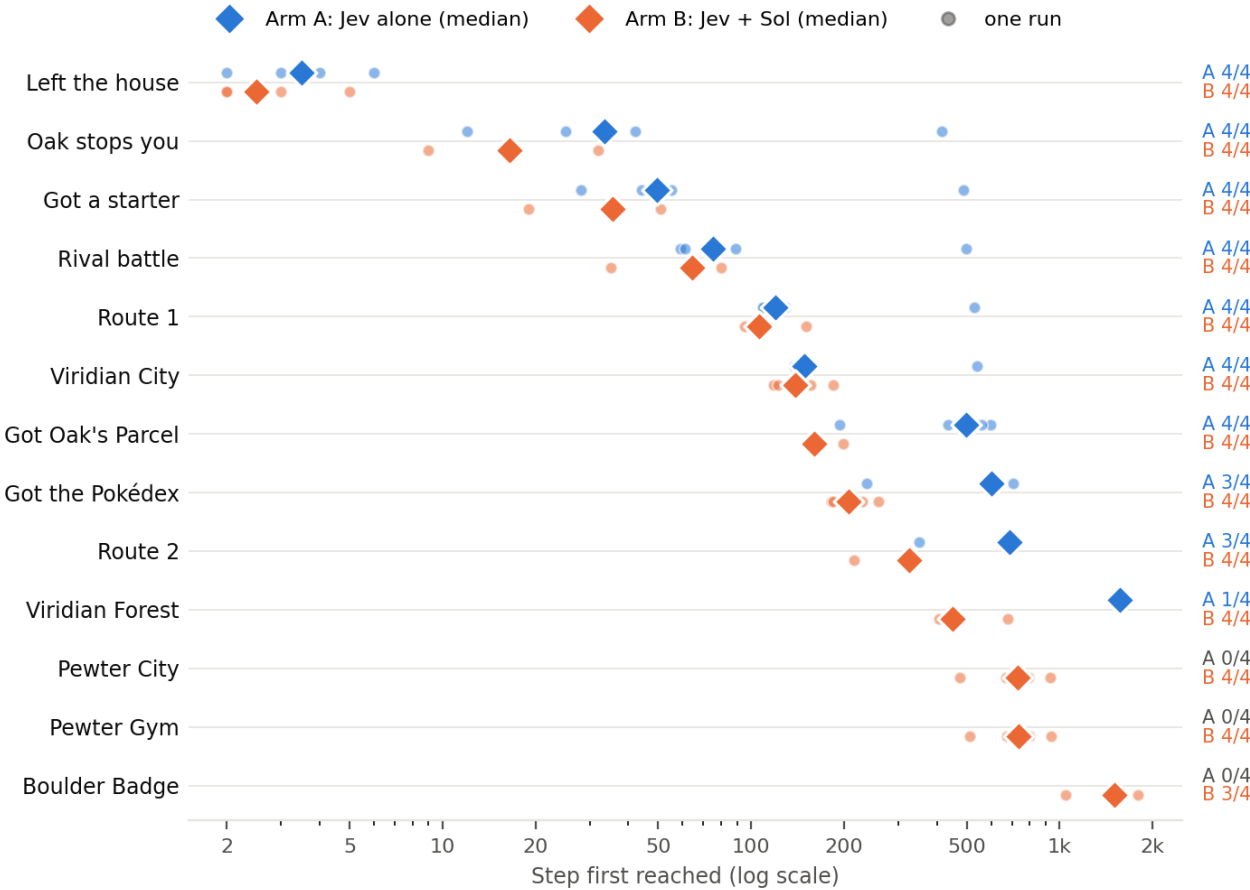
Both arms play from the same new-game save with the same harness. Jev (TypeSafe, via OpenRouter) makes every decision in both arms. In Arm A its goal is fixed (“make progress in the main story toward the next badge”); in Arm B, GPT-6 Sol reads the game state every 15 steps and rewrites that goal. Each run is capped at 2,000 steps and ends early after 1,000 steps without a new milestone.

	Arm A: Jev alone	Arm B: Jev + Sol
Average milestones (of 13)	8.75	12.75
Runs that beat Brock (badge 1)	0 of 4	3 of 4
Average blackouts per run	3.5	6.2
Average cost per run	\$0.08	\$0.61

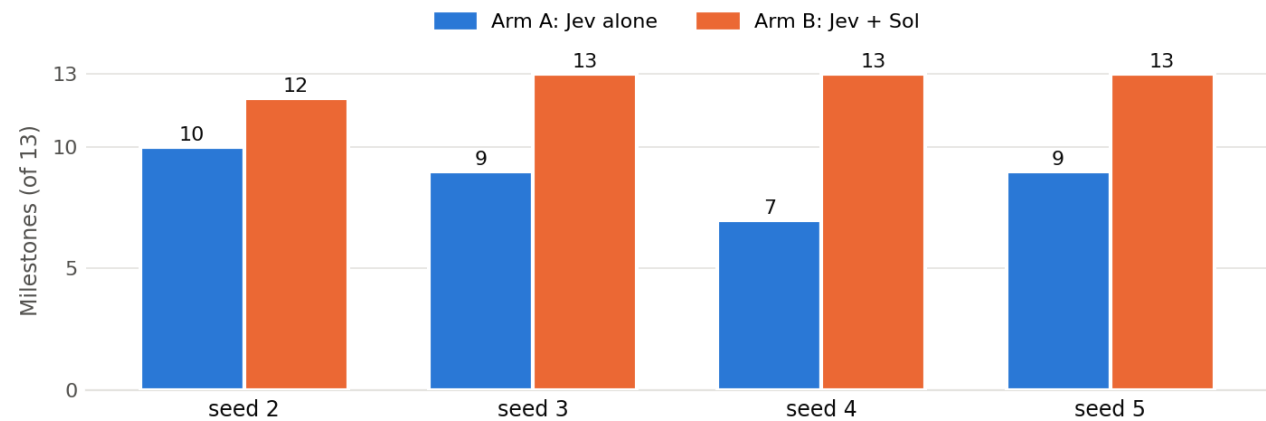
Seeds 2, 3, 4, 5 (4 runs per arm).

How fast each arm reached each milestone

Each small dot is one run; the diamond is the median. The column on the right counts how many runs reached that milestone at all. The step axis is logarithmic so the early milestones stay readable.



Milestones reached per run



Median step per milestone

Median over the runs that reached it; brackets show how many runs did. Speed-up = Arm A median ÷ Arm B median.

Milestone	Arm A median step	Arm B median step	Speed-up
Left the house	3.5 (4/4)	2.5 (4/4)	1.4×
Oak stops you	33.5 (4/4)	16.5 (4/4)	2.0×
Got a starter	49.5 (4/4)	35.5 (4/4)	1.4×
Rival battle	75 (4/4)	64.5 (4/4)	1.2×
Route 1	119.5 (4/4)	106 (4/4)	1.1×
Viridian City	149 (4/4)	139 (4/4)	1.1×
Got Oak's Parcel	496 (4/4)	160 (4/4)	3.1×
Got the Pokédex	602 (3/4)	207 (4/4)	2.9×
Route 2	688 (3/4)	325 (4/4)	2.1×
Viridian Forest	1564 (1/4)	451 (4/4)	3.5×
Pewter City	- (0/4)	733 (4/4)	-
Pewter Gym	- (0/4)	738 (4/4)	-
Boulder Badge	- (0/4)	1507 (3/4)	-

Every run

Arm	Seed	Steps	Score	Starter	Rival	Blackouts	Ember uses	Cost
A	2	2000	10	Bulbasaur	won	4	0	\$0.097
A	3	1722	9	Charmander	lost	4	9	\$0.080
A	4	1558	7	Squirtle	won	2	0	\$0.073
A	5	1350	9	Charmander	lost	4	1	\$0.063
B	2	1511	12	Charmander	won	7	24	\$0.503
B	3	2000	13	Charmander	won	6	29	\$0.630
B	4	2000	13	Squirtle	won	4	0	\$0.655
B	5	2000	13	Charmander	won	8	36	\$0.658

Does the temperature change the result?

The same seeds run at two sampling temperatures, $T = 0.5$ and $T = 1$. Median steps are over the runs that reached each milestone (brackets: how many did).

	Arm A, $T=0.5$	Arm A, $T=1$	Arm B, $T=0.5$	Arm B, $T=1$
Average milestones	7.00	8.75	12.25	12.75
Beat Brock	0 of 4	0 of 4	1 of 4	3 of 4
Left the house	4 (4/4)	3.5 (4/4)	2 (4/4)	2.5 (4/4)
Oak stops you	9 (3/4)	33.5 (4/4)	10.5 (4/4)	16.5 (4/4)
Got a starter	320 (3/4)	49.5 (4/4)	20 (4/4)	35.5 (4/4)
Rival battle	341 (3/4)	75 (4/4)	50.5 (4/4)	64.5 (4/4)
Route 1	606 (3/4)	119.5 (4/4)	84.5 (4/4)	106 (4/4)
Viridian City	616 (3/4)	149 (4/4)	111.5 (4/4)	139 (4/4)
Got Oak's Parcel	642 (3/4)	496 (4/4)	152.5 (4/4)	160 (4/4)
Got the Pokédex	791 (3/4)	602 (3/4)	200.5 (4/4)	207 (4/4)
Route 2	912 (3/4)	688 (3/4)	241.5 (4/4)	325 (4/4)
Viridian Forest	- (0/4)	1564 (1/4)	368 (4/4)	451 (4/4)
Pewter City	- (0/4)	- (0/4)	548.5 (4/4)	733 (4/4)
Pewter Gym	- (0/4)	- (0/4)	652 (4/4)	738 (4/4)
Boulder Badge	- (0/4)	- (0/4)	1527 (1/4)	1507 (3/4)
Average cost per run	\$0.09	\$0.08	\$0.62	\$0.61

What the results say (temperature 1)

This batch repeats the first comparison with one change: Jev's options are now picked at exactly its own probabilities (temperature 1) instead of being sharpened towards its favourite (temperature 0.5). Same seeds (2–5), same 2,000-step cap, same harness.

The planner still wins clearly. Arm B averaged 12.75 of 13 milestones against 8.75 for Arm A. All four Arm B runs reached Pewter City and the gym, and three of the four beat Brock for the Boulder Badge (at steps 1,047, 1,507 and 1,791). No Arm A run reached Pewter City; its furthest was Viridian Forest in one run.

The early gap mostly disappears; the later gap does not. Choosing a starter took Arm A a median of 49.5 steps against 35.5 for Arm B (1.4× slower), compared with 16× slower at temperature 0.5. At temperature 1 Jev's small chance of walking to a Poké Ball is no longer squashed, so it escapes the "keep talking to Oak" loop quickly on its own. But further into the story the planner matters again: Oak's Parcel 3.1× faster in Arm B, the Pokédex 2.9× faster, Route 2 2.1× faster, and only Arm B reliably reaches the forest, Pewter City and the gym.

Temperature 1 is better for both arms. Arm A's average rose from 7.00 to 8.75 and every Arm A run got a starter (at temperature 0.5 one never did). Arm B's average rose from 12.25 to 12.75 and its Brock wins went from 1 of 4 to 3 of 4. Keeping Jev's own probabilities lets it break out of loops, which helps a planner-guided Jev as well as a lone one.

Behaviour worth noting

Starters vary more. At temperature 1 Arm A picked Bulbasaur, Charmander, Squirtle and Charmander across its four runs; Arm B picked Squirtle once (seed 4), which is strong against Brock's rock types and that run won the badge fastest (step 1,047).

Arm B won every rival battle at temperature 1 (4 of 4); Arm A won 2 of 4.

Fewer blackouts for Arm B (6.2 per run against 9.2 at temperature 0.5), even though it spent longer in the hardest areas. Arm A still almost never uses Ember, but did in one run (9 times).

Cost is unchanged: about \$0.08 per Arm A run and \$0.61 per Arm B run.

What this means for the benchmark

The headline conclusion — a slow LLM planner makes a fast decision model get much further through the main story — holds at both temperatures, so it is not an artefact of how Jev's answers are sampled. But the size of the early-game gap depends heavily on temperature, so any single "N× faster" figure should be quoted together with the temperature used. Temperature 1 has no tuned parameter and gave better results for both arms, so it is the better default for future runs.

Caveats

Four runs per arm per temperature is still a small sample. Runs end after 1,000 steps without a new milestone, which may cut Arm A off early.